Functions, Modulus and Quadratics

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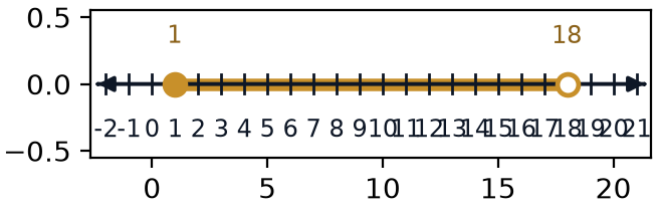
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

[Functions, Modulus & Quadratic Theory · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

2 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

[Differentiation, Rules & Rates of Change · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.



3 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

[Integration, Areas & Kinematics · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

4 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

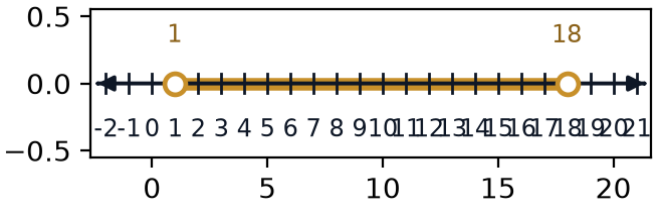
[Logarithms, Exponentials & Straight-Line Reduction · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

5 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

[Permutations & Combinations · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

6 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

[Polynomials, Remainder Theorem & Binomial Theorem · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.



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7 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

[Trigonometry & Circular Measure · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

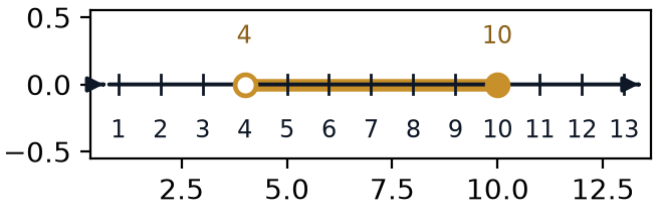
[Vectors in Two Dimensions & Relative Motion · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

9 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

[Functions: Modulus & Inverse · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

10 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

[Indices, Surds & Quadratics · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.



11 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

[Functions: Composite & Inverse · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

12 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

[Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

13 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

[Rational Functions, Limits & Continuity · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

14 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

Given $f(x) = 2x - 3$ for $x \in \mathbb{R}$, sketch $y = |f(x)|$ and solve $|2x - 3| = 5$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Functions, Modulus & Quadratics Practice Exercises** • **Cross-Lesson: 2.01 Simultaneous Equations and Inequalities**

Solve the quadratic equation $2x^2 - 5x - 3 = 0$ by completing the square. State the vertex of the corresponding parabola $y = 2x^2 - 5x - 3$.

16 **[Challenging]** • **Target: Functions, Modulus & Quadratics Practice Exercises** • **Cross-Lesson: 3.01 Indices, Surds and Polynomials**

Find the set of values of the constant k for which the equation $kx^2 + 4x + (k - 3) = 0$ has real roots.

17 **[Challenging]** • **Target: Functions, Modulus & Quadratics Practice Exercises** • **Cross-Lesson: 4.01 Logarithmic and Exponential Functions**

Determine the slant asymptote of $g(x) = \frac{2x^3 - 3x^2 + 5}{x^2 + 1}$ and find the coordinates of any point where the curve intersects this asymptote.

18 **[Challenging]** • **Target: Functions, Modulus & Quadratics Practice Exercises** • **Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure**

Find all vertical and horizontal asymptotes, the exact coordinates of any holes, and all intercepts of the rational function $R(x) = \frac{2x^2 + 5x - 3}{x^2 + 4x + 3}$.

19 **[Challenging]** • **Target: Functions, Modulus & Quadratics Practice Exercises** • **Cross-Lesson: 6.01 Permutations and Combinations**

The function f is defined by $f(x) = x^2 - 6x + 13$ for $x \geq 3$. (a) State the range of f. (b) Find an expression for $f^{-1}(x)$ and state the domain and range of f^{-1}.

20 **[Challenging]** • **Target: Functions, Modulus & Quadratics Practice Exercises** • **Cross-Lesson: 7.01 Differentiation and Rates of Change**

Functions g and h are defined by $g(x) = 2x - 1$ for $x \in \mathbb{R}$, and $h(x) = \frac{6}{x - 2}$ for $xeq2$. (a) Find an expression for $hg(x)$ in simplest form. (b) Solve the equation $hg(x) = g(x)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the modulus equation \(|2x + 3| = 7 \).

a. \(x = 3 \) and \(x = -3 \)

b. \(x = -2 \) and \(x = -\frac{10}{2} \)

c. \(x = 2 \) only

d. \(x = 2 \) and \(x = -\frac{10}{2} \)

22 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

The function $f(x) = \frac{3x+1}{x-2}$ is defined for $xe2$. Find the inverse function $f^{-1}(x)$.

a. $f^{-1}(x) = \frac{3x-2}{x+1}$

b. $f^{-1}(x) = \frac{2x-1}{x-3}$

c. $f^{-1}(x) = \frac{x-2}{3x+1}$

d. $f^{-1}(x) = \frac{2x+1}{x-3} \quad (xe3)$

23 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Find the set of values of the constant m for which the equation $|2x - 6| = mx + 2$ has exactly two distinct real solutions.

a. $-\frac{2}{3} < m < 2$

b. $-2 < m < 2$

c. $m2$

d. $-\frac{2}{3} \le m \le 2$

24 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

Find the set of values of the constant k for which the equation $|x^2 - 4| = 2x + k$ has exactly four distinct real solutions.

a. $4 < k < 5$

b. $4 \le k \le 5$

c. $-4 < k < 4$

d. $k > 5$

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25 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

The functions f and g are defined for all real values of x by $f(x) = 2x - 1$ and $g(x) = \frac{x+5}{3}$. Find the value of x such that $fg(x) = 9$.

a. $x = 10$

b. $x = \frac{23}{2}$

c. $x = 9$

d. $x = 5$

26 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

The quadratic equation $x^2 - (2k+1)x + (k^2+2) = 0$ has real roots α and β. Given that $\alpha^2 + \beta^2 = 27$, find the value of the constant k.

a. 3

b. -5

c. 3 or -5

d. 5

27 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

The line $y = mx - 1$ intersects the curve $y = x^2 + 3x + k$ at two distinct points for all real values of m. Find the range of values of the constant k.

a. $k < -1$

b. $k > -1$

c. $k < 1$

d. $k > 1$

28 [Foundational] • Target: Functions, Modulus & Quadratics Practice Exercises

The function f is defined for $x \in \mathbb{R}$ by $f(x) = x^2 - 2kx + (3k + 10)$, where k is a real constant, and the equation $f(x) = 0$ has no real roots. The function g is defined for $x \in \mathbb{R}$ by $g(x) = |x - k|$. Find the set of values of k for which the minimum value of the composite function $h(x) = f(g(x))$ is strictly greater than 4.

a. $-2 < k < \frac{3+\sqrt{33}}{2}$

b. $\frac{3-\sqrt{33}}{2} < k < \frac{3+\sqrt{33}}{2}$

c. $-2 < k < 5$

d. $0 \leq k < \frac{3+\sqrt{33}}{2}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

The function f is defined for $x \neq 3$ by $f(x) = \frac{2x+1}{x-3}$. Find an expression for $f^{-1}(x)$.

a. $f^{-1}(x) = \frac{3x+1}{x-2}$

b. $f^{-1}(x) = \frac{x-3}{2x+1}$

c. $f^{-1}(x) = \frac{3x-1}{x+2}$

d. $f^{-1}(x) = \frac{3x+1}{x+2}$

30 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

The function f is defined by $f(x) = \frac{2x-1}{x+3}$, for $x \neq -3$. Which of the following expressions represents the inverse function $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{3x+1}{2-x}$, $x \neq 2$

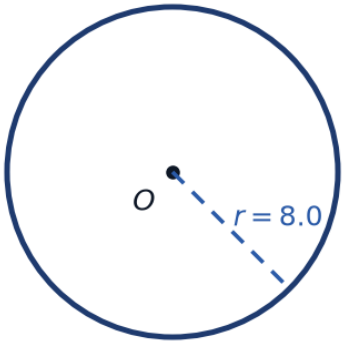
b. $f^{-1}(x) = \frac{x+3}{2x-1}$, $x \neq \frac{1}{2}$

c. $f^{-1}(x) = \frac{3x-1}{x-2}$, $x \neq 2$

d. $f^{-1}(x) = \frac{2x+1}{x-3}$, $x \neq 3$

31 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

A sector of a circle of radius $r = 8$ cm has central angle $\theta = 0.9$ rad. What is the area of the sector?



a. 28.8 cm^2

b. 57.6 cm^2

c. 33.8 cm^2

d. 7.2 cm^2

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32 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the exponential equation $2^{x+2} = 5^x$, giving x in terms of common logarithms.

a. $x = \frac{2\ln 2}{\ln 5 - \ln 2}$

b. $x = \frac{2\ln 5}{\ln 5 - \ln 2}$

c. $x = \frac{\ln 2}{2(\ln 5 - \ln 2)}$

d. $x = \frac{2}{\ln 5 - \ln 2}$

33 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 3$. What is the value of x?

a. 12

b. 3

c. 6

d. 14

34 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

Find all solutions of $2\cos^2 x + 3\sin x - 3 = 0$ in the interval $0 \le x \le 2\pi$.

a. $x = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{\pi}{2}$

b. $x = \frac{\pi}{3},\ \frac{2\pi}{3}$

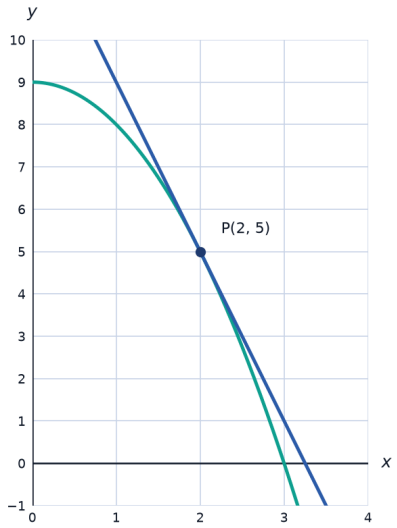
c. $x = \frac{\pi}{6},\ \frac{7\pi}{6}$

d. $x = 0,\ \pi$

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35 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

The diagram shows part of the curve $y = 9 - x^2$ for $x \geq 0$ and the tangent to the curve at the point $P(2, 5)$. Find the exact area of the finite region bounded by the curve, the tangent at P, and the x-axis.



a. $\frac{11}{24}$

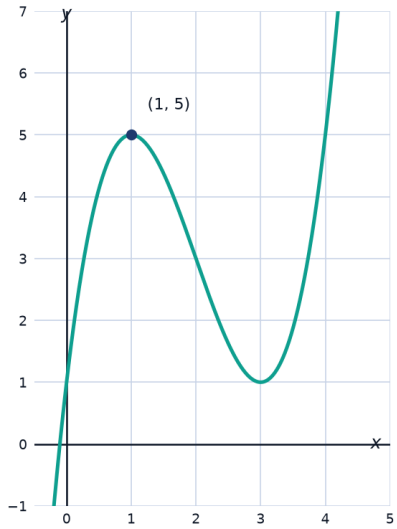
b. $\frac{125}{192}$

c. $\frac{8}{3}$

d. $\frac{25}{8}$

36 [Intermediate] • Target: Functions, Modulus & Quadratics Practice Exercises

The diagram shows part of the curve $y = ax^3 + bx^2 + 9x + 1$, where a and b are constants. The curve has a stationary point at $(1, 5)$. Find the coordinates of the other stationary point on the curve.



a. $(3, 1)$

b. $(3, 0)$

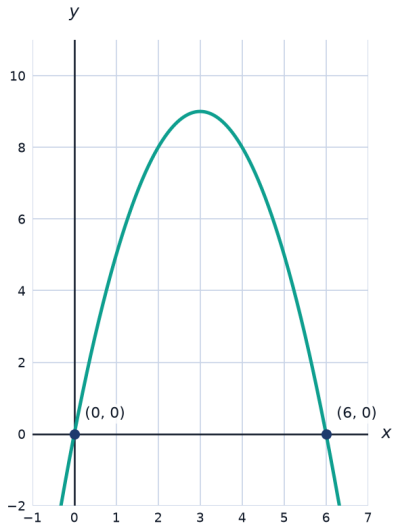
c. $(3, -1)$

d. $(-3, 1)$

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37 **[Intermediate]** • **Target: Functions, Modulus & Quadratics Practice Exercises**

The diagram shows part of the curve $y = 6x - x^2$, which intersects the x-axis at $(0, 0)$ and $(6, 0)$. Find the area of the region enclosed between the curve and the x-axis.



a. 27

b. 36

c. 54

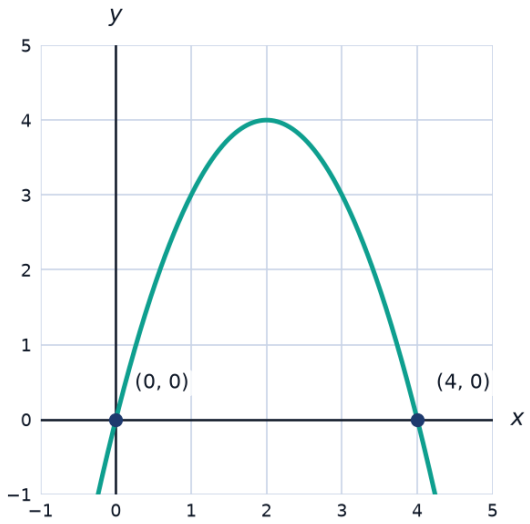
d. 72

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

The diagram shows the curve $y = 4x - x^2$. Find the area of the region enclosed by the curve and the x-axis.



a. $\frac{32}{3}$

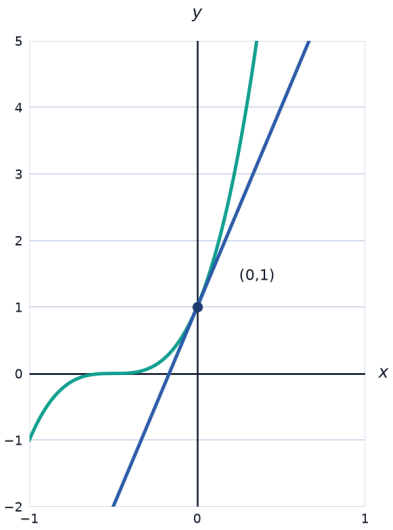
b. 16

c. $\frac{64}{3}$

d. $\frac{16}{3}$

39 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 6.01 Permutations and Combinations

Find the equation of the tangent line to the curve $y = (2x + 1)^3$ at the point where $x = 0$.



a. $y = 2x + 1$

b. $y = 6x - 1$

c. $y = 6x + 1$

d. $y = 6x + 3$

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40 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 7.01 Differentiation and Rates of Change

Two circles, C_1 with centre O_1 and radius 6 cm, and C_2 with centre O_2 and radius $6\sqrt{3}$ cm, intersect at the points A and B. The distance between their centres is $O_1O_2 = 12$ cm. Find the exact area of the region common to both circles, in cm^2.

a. $30\pi - 36\sqrt{3}$

b. $30\pi + 36\sqrt{3}$

c. $42\pi - 36\sqrt{3}$

d. $15\pi - 18\sqrt{3}$

41 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 8.01 Integration and Kinematics

The polynomial $P(x)$ is defined by $P(x) = (2 + ax)^n + bx^4 + cx - 4$ where n is an integer greater than 2, and a, b, c are constants with $a > 0$. In the binomial expansion of $(2 + ax)^n$ in ascending powers of x: - the coefficient of x^2 is twice the coefficient of x, - the coefficient of x^3 is equal to the coefficient of x^2. Given that $(x + 2)$ is a factor of $P(x)$, and that the remainder when $P(x)$ is divided by $(x + 1)$ is 7, find the remainder when $P(x)$ is divided by $(x - 1)$.

a. 1011

b. 1031

c. 1015

d. 499

42 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 2.01 Simultaneous Equations and Inequalities

At 08:00, a cargo vessel has position vector $(-7\mathbf{i} + 22\mathbf{j})$ km relative to a coastal radar station at the origin O, and moves with a constant velocity of $(15\mathbf{i} + 5\mathbf{j})$ km/h. At the same time (08:00), a high-speed patrol boat stationed at $(5\mathbf{i} + 6\mathbf{j})$ km sets off in a straight line at a constant speed v km/h to intercept the cargo vessel. Find the minimum speed v required for the patrol boat to intercept the cargo vessel, and the time at which interception occurs at this minimum speed.

a. Minimum speed 15 km/h, interception at 12:00

b. Minimum speed 15 km/h, interception at 10:00

c. Minimum speed $5\sqrt{10}$ km/h, interception at 12:00

d. Minimum speed 12.5 km/h, interception at 11:12

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43 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 3.01 Indices, Surds and Polynomials

Six-digit numbers are formed using digits chosen from the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$, with no digit repeated in any number. Find the number of different six-digit odd numbers that can be formed such that the digits 2 and 4 are not adjacent.

a. 8160

b. 7680

c. 9120

d. 2880

44 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 4.01 Logarithmic and Exponential Functions

The real numbers x and y satisfy the simultaneous equations: $\log_3(x) + 2\log_9(y-1) = 2$ $4^y \cdot 8^{-\frac{x}{3}} = 32$ Find the value of $x^2 + y^2$.

a. 25

b. 10

c. $\frac{145}{4}$

d. 50

45 [Challenging] • Target: Functions, Modulus & Quadratics Practice Exercises • Cross-Lesson: 5.01 Trigonometric Functions and Circular Measure

A supply launch S departs from a base at the origin O at time $t = 0$ to intercept an automated cargo barge C. At $t = 0$, the barge has position vector $\mathbf{r}_0 = (16; 12)$ km relative to O and maintains a constant velocity of $\mathbf{v}_C = (-15; 0)$ km/h. The launch travels in a straight line with constant velocity $\mathbf{v}_S$ and constant speed v km/h. Determine the minimum speed v required for the launch to successfully intercept the barge.

a. 9 km/h

b. 12 km/h

c. 15 km/h

d. 7.2 km/h

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